Project 1

ArtBench Generative Modeling

Generative AI | 2025/2026 | 2nd Semester

João Correia · Penousal Machado · Pedro Louro · Sancho Simões

Deadline: 12 April 2026

What Is This Project About?

Train generative models on artwork images and produce new plausible samples that match the data distribution.

The Task

- Learn to generate images of paintings
- Multiple artistic styles (10 classes)
- Balance fidelity, diversity, and style

The Challenge

- Compare different model families
- Rigorous quantitative evaluation
- Statistical reporting across seeds

This is NOT classification — there is no single right answer. Quality is measured by distribution-level metrics.

Groups of 2 students · Individual grading

Dataset: ArtBench-10

A curated dataset of artworks designed to benchmark generative models in the visual arts domain.

10
Artistic Styles

50K
Training Images

32×32
RGB Resolution

Development Phase

- Use the provided 20% subset
- Faster iteration on architectures
- Explore hyperparameters and strategies

Final Evaluation

- Retrain best model on full dataset
- 50,000 training samples
- Final metrics computed here

Required Model Families

You must implement, train, and compare at least one model from each of these three families.

Autoencoders

At least one VAE-type model

Strengths:

- Stable training
- Smooth latent space

Weakness:

- Blurry outputs

GANs

At least one DCGAN model

Strengths:

- Sharp samples
- Fast generation

Weakness:

- Unstable training

Diffusion

At least one diffusion model

Strengths:

- High quality
- Good diversity

Weakness:

- Slow inference

Additional methods beyond these three are encouraged and may earn extra credit.

Evaluation Protocol

1. Generate 5,000 samples per model
2. Sample 5,000 real images from test split
3. Compute FID and KID
4. Repeat 10× with different random seeds
5. Report mean ± standard deviation

FID — Fréchet Inception Distance

- Full 5,000 generated vs 5,000 real
- Lower = better quality + diversity

KID — Kernel Inception Distance

- 50 subsets of size 100
- Report mean ± std across subsets

Min 10 seeds per config · Same preprocessing and eval code across all models · Include visual sample grids

Your Pipeline

Build a complete generative pipeline from data loading to evaluation.

Development Strategy

1. Start with the 20% subset for fast iteration
2. Compare all three model families
3. Select best approach
4. Retrain on full 50K dataset
5. Final evaluation with 10 seeds

Starting Resources

- Starter codebase provided
- ArtBench paper (arXiv:2206.11404)
- HuggingFace diffusers tutorial
- PyTorch GAN Zoo (Facebook)
- Reference packs for each family

Report & Submission

Report Structure

- Introduction
- Problem description
- Methodology and approach
- Experimental setup
- Analysis of results
- Conclusion

Format & Rules

- Springer LNCS format (LaTeX)
- Max 12 pages (excl. references)
- Portuguese or English
- Submit via Inforestudante
- Overleaf template available

What We Look For

- Clear description of adopted approach
- Model architectures and training details
- Reproducible protocol (hyperparams, seeds)
- Quantitative evaluation with proper metrics

AI Tool Policy

- You may use AI tools (code assistants, LLMs)
- All AI use must be explicitly acknowledged
- You are fully responsible for correctness

Summary & Key Dates

Deadline: 19 April 2026

Submit via Inforestudante

- ✓ Train at least 3 model families: VAE, GAN, Diffusion
- ✓ Develop on 20% subset, final eval on full dataset
- ✓ Compute FID and KID with 10 random seeds
- ✓ Write a 12-page report in Springer LNCS format
- ✓ Acknowledge all sources including AI tools
- ✓ Public defence — individual grading

Good luck!